

Report 5: Gaussian Blur

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1 Implementation

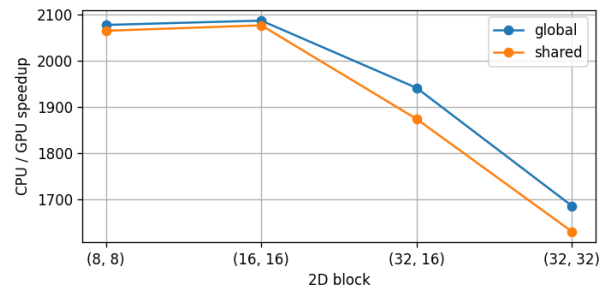
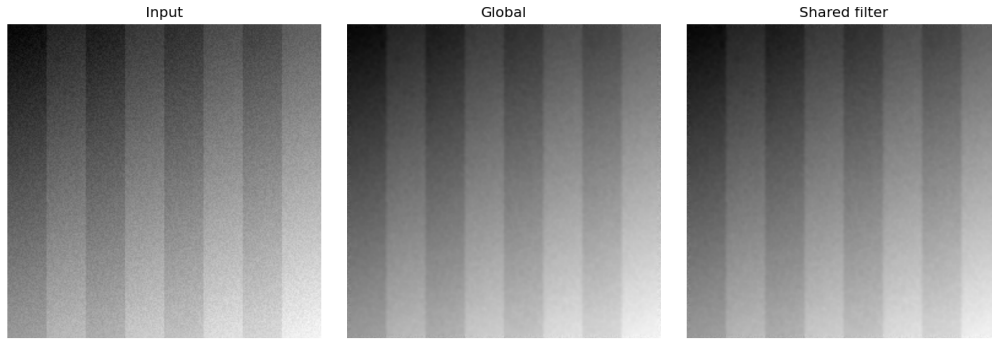
I applied a 7×7 Gaussian filter to the grayscale image. The coefficients sum to 1003. One kernel reads the filter from global memory; the other copies its 49 coefficients into shared memory before convolution. Both preserve the three-pixel border.

I ran the notebook on Google Colab with a Tesla T4 and a 256×256 sample image. GPU timings use `time.time()` after warm-up and synchronization, averaged over three runs; allocation and image transfers are excluded. Where measured, CPU time is one run; speedup is relative to that Python CPU implementation.

2 Results

CPU time: 404.6767 ms. Maximum pixel error: 0.

Block	Global (ms)	Shared filter (ms)
(8, 8)	0.1947	0.1959
(16, 16)	0.1938	0.1948
(32, 16)	0.2084	0.2158
(32, 32)	0.2399	0.2481



3 Discussion

The 16×16 block gave the best time for both versions. Shared filter memory was slightly slower for every tested block. Its loading and synchronization overhead may outweigh its benefit on this small image.